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Ionic conductivity of Li^+ ion conductors $Li_2M^{3+}M^{4+}P_3O_{12}$

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Abstract

Lithium ion conductors based on $Li_2M^{3+}M^{4+}P_3O_{12}$ (M^{3+} = Cr, Fe, In; M^{4+} = Ti, Zr and Hf) have been synthesized. The $Li_2InTiP_3O_{12}$ show larger conductivity ($\sigma_{573\ K}$ is 1.22×10^{-3} S cm^{-1}) in the series of compounds under study and the results are discussed in terms of the polarizability of the network cation and the unit cell volume.

Keywords: Ionic conductivity; Lithium; NASICON; Phosphates

1. Introduction

The low equivalent weight and the high electropositive nature of Li which gives high cell voltage and energy density of advanced electrochemical devices, paved the way for an intense search for a good Li^+ ion conductor with the requisite ceramic properties over the past several years. The conductivity of the compound $Li_3Zr_2Si_2PO_{12}$, analogous to NASICON, is three orders of magnitude lower than the conductivity of the latter due to the larger tunnel size [1]. As the structure is amenable to extensive crystal chemical substitutions retaining the basic structure, by suitable modification of the framework cations, the tunnel size for Li ion migration can be modified and fast lithium ion conductors can be achieved [2–4].

Previous studies have established that substitution of a trivalent cation at the Ti site increases the conductivity of $LiTi_2P_3O_{12}$. The unit cell remains rhombohedral with the space group R$\bar{3}$c in the solid solution $Li_{1+x}Ti_{2-x}M_xP_3O_{12}$ (where M = Cr^{3+} and In^{3+}) upto $x = 0.8$ for Cr and $x = 0.35$ for In and these compositions show a maximum conductivity [5,6]. Substitutional studies in the $Li_{1+x}Hf_{2-x}In_xP_3O_{12}$ system have indicated that the rhombohedral phase exists upto $x = 0.3$ [7]. From all the above studies it is evident that the value of x at which the crystal symmetry changes from rhombohedral to orthorhombic is dependent on the size of the tri/tetravalent cation at the octahedral site. Hence, to compare the nature of the conduction in a series of compounds with various tri/tetravalent cations, the x composition in the present study is chosen to be 1, i.e. $Li_2M^{3+}M^{4+}P_3O_{12}$ so that all the compositions have the same crystal symmetry, viz. orthorhombic.

2. Experimental

The compounds were prepared by high temperature solid state reaction starting from the respective oxides/carbonates and ammonium dihydrogen phosphate. The final sintering temperature was 1100–1200°C. To avoid the loss of Li, the pellets were

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wrapped in a Pt foil and placed in an evacuated, sealed quartz tube and heated. Phase purity is established by powder X-ray diffraction (Cu Kα_1 radiation) and the unit cell parameters are computed by the LSQ fit of the high angle reflections. Ionic conductivity measurements are carried out on the samples in the form of pellets which were polished with a 400 grade emery paper and Pt paint was coated on the surface and the pellets were cured at 600°C for 6 h to remove the volatile organic matter. Impedance measurements using Pt electrodes are carried out in the frequency range 100 Hz–15 MHz between 300 K to 873 K in Ar atmosphere on pellets from the same batch and from different batches to check reproducibility of the results.

3. Results and discussion

3.1. Phase identification

The compounds $Li_2M^{3+}M^{4+}P_3O_{12}$ are found to be single phasic in nature crystallizing in orthorhombic structure with the space group Pbca. The lattice parameters are listed in Table 1. Our results are in agreement with the crystal structure data reported by Hamdoune et al., in $Li_{1+x}Ti_{2-x}In_xP_3O_{12}$ [8]. The lattice parameters increase with increasing ionic radius of the M^{3+}/M^{4+} ion at the octahedral site (e.g., Cr^{3+}, Fe^{3+}, In^{3+}; Ti^{4+}, Zr^{4+}, Hf^{4+}). The X-ray patterns are complex in the case of the Hf compounds due to the lowering of symmetry (presumably monoclinic). The lattice parameters of the Zr and Hf analogues are of the same order due to the similar sizes of the Zr^{4+} and Hf^{4+}.

Table 1
Lattice parameters of $Li_2M^{3+}M^{4+}P_3O_{12}$

Compound	a, Å	b, Å	c, Å	V, Å^3
$Li_2CrTiP_3O_{12}$	8.467	8.688	23.83	1752
$Li_2CrZrP_3O_{12}$	8.596	8.787	24.21	1828
$Li_2CrHfP_3O_{12}$	8.598	8.816	24.21	1835
$Li_2FeTiP_3O_{12}$	8.452	8.672	23.62	1731
$Li_2FeZrP_3O_{12}$	8.470	8.720	23.80	1758
$Li_2InTiP_3O_{12}$	8.577	8.826	24.05	1821
$Li_2InZrP_3O_{12}$	8.767	8.979	24.20	1905
$Li_2InHfP_3O_{12}$	8.757	8.954	24.28	1904

3.2. Conductivity measurements

Ionic conductivity results of all the phases obtained from complex impedance plots are analyzed by the Arrhenius equation, $\sigma T=\sigma_0\exp(-E_a/kT)$, where σ_0 and E_a represent the pre-exponential factor and the activation energy respectively. An estimate of the grain interior and grain boundary conductivities are made by resolving the impedance spectra by means of a software package [9]. The σ_{RT}, $\sigma_{573\ K}$, σ_0 and E_a values of the phases studied are listed in Table 2. The maximum dispersion in the E_a values obtained from measurements on different pellets of a given composition is 1 to 1.5 kJ/mol. Fig. 1 illustrates the linear plots of log(σT) vs. 1000/T for the Cr and In analogues. The $Li_2InTiP_3O_{12}$ and $Li_2InZrP_3O_{12}$ compounds show that the Arrhenius plots have two straight lines with differing slopes. The low temperature region has a slope of 34 kJ/mol and 58 kJ/mol and at the high temperature region 21 and 48 kJ/mol for the Ti and Zr analogues respectively. The E_a value of $Li_2InTiP_3O_{12}$ compares well

Table 2
Conductivity data for $Li_2M^{3+}M^{4+}P_3O_{12}$

Compound	σ_{RT}, S cm^{-1}	$\sigma_{573\ K}$, S cm^{-1}	σ_0, S cm^{-1}K	E_a, kJ/mol
$Li_2CrTiP_3O_{12}$	4.41×10^{-7}	5.56×10^{-4}	2.70×10^{4}	54.03
$Li_2CrZrP_3O_{12}$	5.91×10^{-7}	5.92×10^{-4}	3.94×10^{4}	55.00
$Li_2CrHfP_3O_{12}$	4.20×10^{-8}	2.74×10^{-4}	4.89×10^{4}	60.79
$Li_2FeTiP_3O_{12}$	3.34×10^{-7}	5.43×10^{-4}	9.94×10^{4}	59.82
$Li_2FeZrP_3O_{12}$	1.44×10^{-7}	3.82×10^{-4}	8.67×10^{4}	61.75
$Li_2InTiP_3O_{12}$	8.30×10^{-6}	1.22×10^{-3}	2.03×10^{1}	21.23
			3.99×10^{1}	33.77
$Li_2InZrP_3O_{12}$	8.22×10^{-8}	1.44×10^{-4}	2.72×10^{3}	48.25
			1.50×10^{4}	57.89
$Li_2InHfP_3O_{12}$	2.74×10^{-8}	2.52×10^{-5}	7.33×10^{3}	61.75

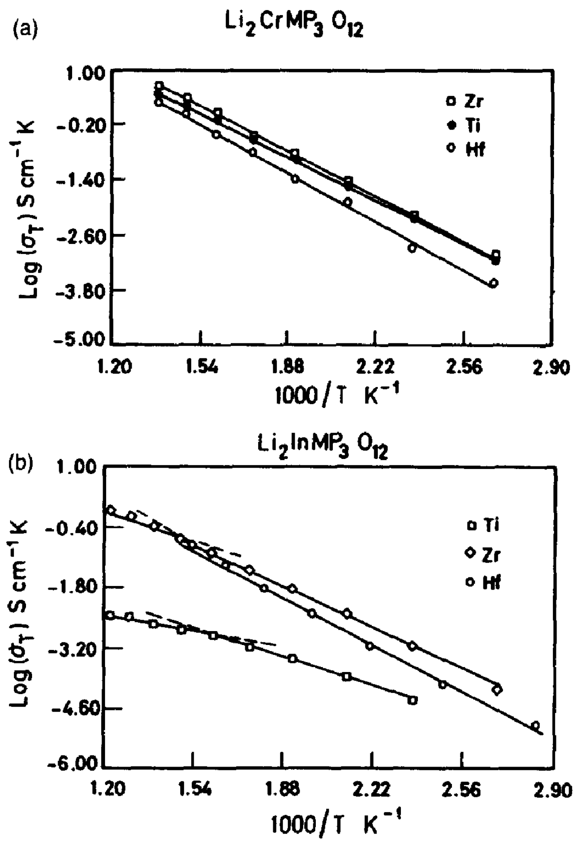

Fig. 1. Arrhenius plots for (a) $Li_2CrMP_3O_{12}$ and (b) $Li_2InMP_3O_{12}$.

with that reported in the literature [5,6]. The break in the slope is attributed to a phase transformation to a more conducting phase at higher temperatures. This phase transformation has been confirmed by the endothermic hump in the DSC peaks in both the samples. The Cr analogues on the other hand, have a single activation energy throughout the temperature range of measurement.

Some general trends are evident from the data presented in Table 2. The activation energy increases with increasing radius of the tetravalent cation. The Hf analogues have lower conductivity. This is due to a large unit cell volume and consequently bigger bottleneck size. It is known that in the NZP system of compounds, Na^+ containing compounds have maximum ionic conductivity, e.g. NASICON and the low conductivity of Li^+ in the structure type is attributed to the 'rattling of Li^+ ions' within the large interstitial spaces available [10]. The conductivity of Li^+ in the $Li_2InTiP_3O_{12}$ is the highest in the present study. This is due to the low ionicity and high polarizability of In^{3+} and probably optimum bottleneck size. The larger the polarizability of the framework cation and the lower the ionicity of the M–O bond, the easier the transport of the mobile ions (e.g. Li^+) in the tunnels.

The log f vs. σ plots are shown in Fig. 2a and 2b. All the phases show frequency-independent plateau corresponding to dc conductivity. At high temperatures, low frequency dispersion is observed due to incipient electrode polarisation effects. At these temperatures, the ionic conductivity is high enough to produce a significant build-up of charges at the electrodes, which reduces the effective applied field across the sample and hence the apparent conductivity. At high frequencies the period of the applied field is too short for the charging to occur and the ac conductivity is generally taken to assume the fre-

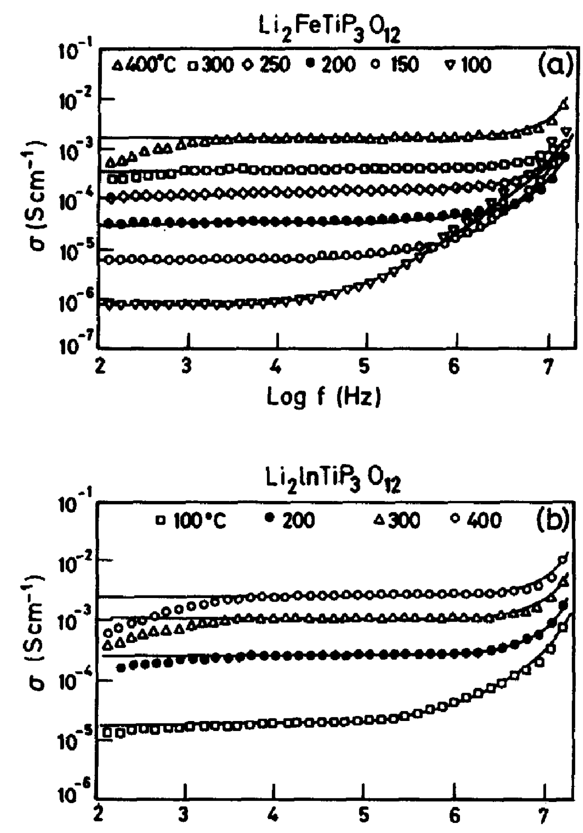

Fig. 2. Conductivity (σ) vs. frequency (log f) plot at various temperatures for (a) $Li_2FeTiP_3O_{12}$ (b) $Li_2InTiP_3O_{12}$; solid lines indicate fit to the power law (see Section 3.2).

quency independent value, which is equal to the true dc conductivity. At higher frequencies the conductivity dispersion follows the power law dependence, as expected for most of the hopping type ionic conductors [11], $\sigma(\omega) = \sigma_0 + A\omega^n$, where σ_0 is a dc term, A is a temperature dependant parameter and n is the slope of the high frequency dispersion data and is less than 1. This high frequency dispersion could be attributed to a cooperative event involving many neighbouring ions. The solid curves in Fig. 2a and 2b indicate the theoretical fit for different temperatures. The attempt frequency ω_p, is given by $\omega_p = [\sigma_0/A]^{1/n}$ and is proportional to ion hopping rate. A plot of log ω_p vs. $1000/T$ for $Li_2CrMP_3O_{12}$ (M = Ti, Zr, Hf) is shown in Fig. 3. The linear dependence indicates that the ω_p is a thermally activated quantity and bears the same activation energy within experimental error. From the infinite temperature intercept, the effective attempt frequencies were calculated. The ω_e values are of the order of 10^{12} which are of the same order of magnitude as that of ω_0 (vibrational frequency of the mobile ion); this indicates that the entropy of the hopping process in the equation

$$\omega_e = \omega_0 \exp(\Delta S_a/k)$$

is small in these materials which is expected for all fast ion conductors [12,13].

σ_0 is given by $K\omega_p$ where the magnitude of the constant K is largely a measure of the mobile ion concentration. It has been found in the present study that the K values range from 10^{-12} to 10^{-9} $\Omega^{-1}\,cm^{-1}\,Hz^{-1}$, depending on the composition, in the measured temperature range, while the value of K for Na β-alumina is of the order of 1.5×10^{-12} $\Omega^{-1}\,cm^{-1}\,Hz^{-1}$ and of the order of 10^{-14} for $LiGaO_2$ [14]. This indicates that the compounds under investigation have relatively high mobile ion concentrations.

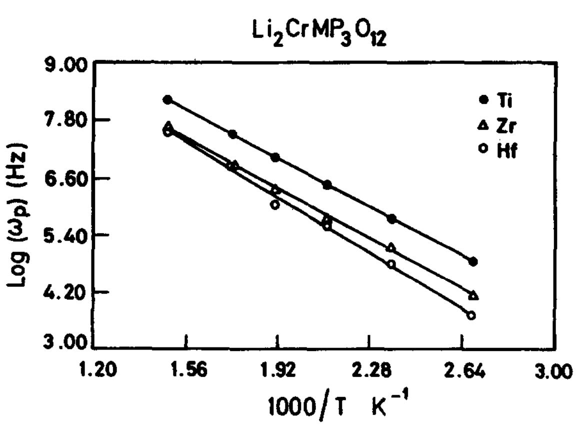

Fig. 3. Arrhenius plot of the attempt frequency (ω_p) for $Li_2CrMP_3O_{12}$.

4. Conclusions

The conductivity of the $LiM_2^{4+}P_3O_{12}$ could be enhanced by the substitution of a trivalent metal ion, thereby increasing the occupancy of the Li ions in the type II sites. The In analogues show more promise towards this end due to the very large polarizability and low ionicity of In^{3+}. Hence, polarizability of the network cation plays a crucial role in improving the ionic conductivity. The present study shows that a critical interplay of several factors are responsible for the increased conductivity in $Li_2M^{3+}M^{4+}P_3O_{12}$ vis a vis the $LiM_2P_3O_{12}$ analogues.

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